

第八章 玻色统计和费米统计

§ 8.1 热力学量的统计表达

经典极限条件 / 非简并条件:

$$e^{-\alpha} = \frac{N}{V} \left(\frac{h^2}{2\pi m k T} \right)^{3/2} \sim n \lambda^3 \ll 1$$

简并条件 $\rightarrow$ 量子统计关联.

系统与热库 / 粒子源 达到热动平衡

$\hookrightarrow$ 温度, 化学势 给定. $\rightarrow$ 确定 α, β .

1. 玻色系统

(1) 平均粒子数: $\bar{N} = \sum_i \bar{n}_i = \sum_i \frac{e^{\alpha + \beta \epsilon_i}}{e^{\alpha + \beta \epsilon_i} - 1}$

引入巨配分函数:

$$\Xi = \sum_i \Xi_i = \sum_i (1 - e^{-\alpha - \beta \epsilon_i})^{-1}$$

$$\ln \Xi = - \sum_i \ln (1 - e^{-\alpha - \beta \epsilon_i})$$

$$\hookrightarrow \bar{N} = \frac{\partial}{\partial \alpha} \ln \Xi$$

(2) 内能 $\bar{E} = \sum_i \epsilon_i \bar{n}_i = \sum_i \frac{\epsilon_i e^{\alpha + \beta \epsilon_i}}{e^{\alpha + \beta \epsilon_i} - 1} = - \frac{\partial}{\partial \beta} \ln \Xi$

(3) $\bar{p} \times V = \sum_i \frac{\partial \epsilon_i}{\partial y} \bar{n}_i = \sum_i \frac{\epsilon_i}{e^{\alpha + \beta \epsilon_i} - 1} \frac{\partial \epsilon_i}{\partial y} = - \frac{1}{\beta} \frac{\partial}{\partial y} \ln \Xi$

例: 对压强 p , $y = -V \Rightarrow p = \frac{1}{\beta} \frac{\partial}{\partial y} \ln \Xi$

(4) 熵 $dS = \frac{1}{T} (dU - \bar{Y} dV - \mu d\bar{N})$

考虑: $\beta \cdot (dU - \bar{Y} dV + \frac{\mu}{T} d\bar{N})$

$$= \beta \left[-\alpha \left(\frac{\partial \ln \Xi}{\partial \beta} \right) + \frac{1}{\beta} \alpha \left(\frac{\partial \ln \Xi}{\partial y} \right) dV - \frac{\mu}{T} \alpha \left(\frac{\partial \ln \Xi}{\partial \alpha} \right) \right]$$

$$= -\beta \alpha \left(\frac{\partial \ln \Xi}{\partial \beta} \right) + \alpha \left(\frac{\partial \ln \Xi}{\partial y} \right) dV - \alpha \alpha \left(\frac{\partial \ln \Xi}{\partial \alpha} \right)$$

$$\Xi = \Xi(\alpha, \beta, y)$$

$$d(\ln \Xi) = \frac{\partial \ln \Xi}{\partial \alpha} d\alpha + \frac{\partial \ln \Xi}{\partial \beta} d\beta + \frac{\partial \ln \Xi}{\partial y} dy$$

$$= -\beta \alpha \left(\frac{\partial \ln \Xi}{\partial \beta} \right) + \alpha \left(\frac{\partial \ln \Xi}{\partial y} \right) dV - \frac{\partial \ln \Xi}{\partial \alpha} d\alpha - \frac{\partial \ln \Xi}{\partial \beta} d\beta - \alpha d \left(\frac{\partial \ln \Xi}{\partial \alpha} \right)$$

$$= d(\ln \Xi - \beta \alpha \left(\frac{\partial \ln \Xi}{\partial \beta} \right) - \alpha \left(\frac{\partial \ln \Xi}{\partial y} \right) dV) \Rightarrow \beta \text{ 为积分因子, } \beta = \frac{1}{kT}, \mu = -\frac{\mu}{T}$$

$$dS = k \cdot \beta (dU - \gamma dV + \frac{\partial}{\partial \beta} dN)$$

$$= k (d(\ln \bar{z}) - \beta \frac{\partial \ln \bar{z}}{\partial \beta} - \alpha \frac{\partial \ln \bar{z}}{\partial \alpha})$$

$$S = k (\ln \bar{z} - \alpha \frac{\partial \ln \bar{z}}{\partial \alpha} - \beta \frac{\partial \ln \bar{z}}{\partial \beta})$$

$$= k (\ln \bar{z} + \alpha \bar{n} + \beta \bar{U})$$

$$= k \left[\sum_i -w_i \ln (1 - e^{-\alpha - \beta \epsilon_i}) + \alpha \bar{n} + \beta \bar{U} \right]$$

$$= k \sum_i (-w_i \ln \frac{e^{\alpha + \beta \epsilon_i} - 1}{e^{\alpha + \beta \epsilon_i}} + \alpha \bar{n} + \beta \bar{U})$$

$$= k \sum_i -w_i \ln \left(\frac{w_i}{\bar{w}_i} \right) + (\bar{w}_i + \bar{U}_i) (\alpha + \beta \epsilon_i)$$

$$= k \sum_i -w_i \ln w_i + w_i \ln \bar{w}_i + (\bar{w}_i + \bar{U}_i) \ln \left(1 + \frac{w_i}{\bar{w}_i} \right)$$

$$= k \sum_i (\bar{w}_i + \bar{U}_i) \ln (\bar{w}_i + \bar{U}_i) - w_i \ln w_i + \bar{U}_i \ln \bar{w}_i$$

$$= k \ln \bar{w} \quad \text{符合玻尔兹曼关系}$$

熵力学势

$$\Phi = U - TS - \bar{U}/\mu$$

$$= -\frac{\partial}{\partial \beta} \ln \bar{z} - T k (\ln \bar{z} - \alpha \frac{\partial \ln \bar{z}}{\partial \alpha} - \beta \frac{\partial \ln \bar{z}}{\partial \beta}) - \frac{\partial}{\partial \alpha} \ln \bar{z} \cdot \frac{1}{\beta}$$

$$= -k T \ln \bar{z}$$

2. 共形映射

引又反配分函数

$$Z := \sum_i \tau_i = \sum_i \tau_i (1 + e^{-\alpha + \beta \varepsilon_i})^{-w_i}$$

$$\ln Z = - \sum_i w_i \ln(1 + e^{-\alpha + \beta \varepsilon_i})$$

(1) 平均粒子数 $\bar{n} = \sum_i \frac{w_i}{e^{\alpha + \beta \varepsilon_i} + 1} = \frac{\partial}{\partial \alpha} \ln Z$

(2) 内能 $U = \sum_i \frac{w_i \varepsilon_i}{e^{\alpha + \beta \varepsilon_i} + 1} = \frac{\partial}{\partial \beta} \ln Z$

(3) 熵力 $\gamma = \sum_i \frac{\partial \varepsilon_i}{\partial y} \frac{w_i}{e^{\alpha + \beta \varepsilon_i} + 1} = \frac{1}{\beta} \frac{\partial}{\partial y} \ln Z$

(4) 熵 $dS = \frac{1}{T} (dU - \gamma dy - \mu d\bar{n})$

考虑: $\beta \cdot (dU - \gamma dy + \frac{1}{\beta} d\bar{n})$

$$= \beta \left[d \left(\frac{\partial}{\partial \beta} \ln Z \right) - \frac{1}{\beta} \frac{\partial}{\partial y} \ln Z dy + \frac{\partial}{\partial \beta} d \left(\frac{\partial}{\partial \alpha} \ln Z \right) \right]$$

$$= \beta d \left(\frac{\partial}{\partial \beta} \ln Z \right) - \frac{\partial}{\partial y} \ln Z dy + d \left(\frac{\partial}{\partial \alpha} \ln Z \right)$$

$$\ln Z = \ln Z(\alpha, \beta, y)$$

$$d(\ln Z) = \frac{\partial}{\partial \alpha} (\ln Z) d\alpha + \frac{\partial}{\partial \beta} (\ln Z) d\beta + \frac{\partial}{\partial y} (\ln Z) dy$$

$$= \beta d \left(\frac{\partial}{\partial \beta} \ln Z \right) - d(\ln Z) + \frac{\partial}{\partial \alpha} (\ln Z) d\alpha + \frac{\partial}{\partial \beta} (\ln Z) d\beta + d \left(\frac{\partial}{\partial \alpha} \ln Z \right)$$

$$= d(-\ln Z + \alpha \frac{\partial}{\partial \alpha} \ln Z + \beta \frac{\partial}{\partial \beta} \ln Z)$$

$$\beta = \frac{1}{kT}$$

$$dS = \frac{1}{T} d(-\ln Z + \alpha \frac{\partial}{\partial \alpha} \ln Z + \beta \frac{\partial}{\partial \beta} \ln Z)$$

$$= k d(-\ln Z + \alpha \frac{\partial}{\partial \alpha} \ln Z + \beta \frac{\partial}{\partial \beta} \ln Z)$$

$$\rightarrow S = k(-\ln Z + \alpha \frac{\partial}{\partial \alpha} \ln Z + \beta \frac{\partial}{\partial \beta} \ln Z)$$

$$= k \sum_i \left[w_i \ln(1 + e^{\alpha + \beta \varepsilon_i}) - \alpha w_i \frac{e^{\alpha + \beta \varepsilon_i}}{1 + e^{\alpha + \beta \varepsilon_i}} - \beta w_i \frac{\varepsilon_i e^{\alpha + \beta \varepsilon_i}}{1 + e^{\alpha + \beta \varepsilon_i}} \right]$$

$$a_i = \frac{w_i}{e^{\alpha + \beta \varepsilon_i} + 1}$$

$$= k \sum_i \left[w_i \ln \left(\frac{w_i}{a_i} \right) - (\alpha + \beta \varepsilon_i) e^{\alpha + \beta \varepsilon_i} a_i \right]$$

$$= k \sum_i \left[w_i \ln w_i - w_i \ln a_i - \ln \left(\frac{w_i}{a_i} - 1 \right) \left(\frac{w_i}{a_i} - 1 \right) a_i \right]$$

$$= k \sum_i \left[w_i \ln w_i - (w_i - a_i) \ln(w_i - a_i) - a_i \ln a_i \right]$$

§ 8.2 弱简并理想玻色气体和费米气体

当 e^{α} , $h\lambda^3$ 为一阶小量时 (弱简并), 气体宏观参量有何变化?

$$\epsilon = \frac{1}{2m}(p_x^2 + p_y^2 + p_z^2)$$

在体积 V 内, 能量分布于 $\epsilon \sim \epsilon + d\epsilon$ 之间的状态数为

$$\begin{aligned} D(\epsilon) d\epsilon &= g \frac{V}{h^3} dp_x dp_y dp_z \quad \text{量子态数} = \text{相空间体积} \\ &= g \frac{4\pi p^2 V}{h^3} dp = g V \frac{4\pi (2m\epsilon)^{3/2}}{h^3} d(\sqrt{2m\epsilon}) \\ &= \frac{4\sqrt{2} V \pi g m^{3/2}}{h^3} \epsilon^{1/2} d\epsilon \end{aligned}$$

$$\begin{aligned} \text{总粒子数 } N &= \sum_i n_i = \sum_{\epsilon} e^{\alpha + \beta \epsilon} D(\epsilon) \\ &= \frac{4\sqrt{2} V \pi g m^{3/2}}{h^3} \int_0^\infty \frac{\epsilon^{1/2}}{e^{\alpha + \beta \epsilon} \pm 1} d\epsilon \end{aligned}$$

$$\begin{aligned} \text{内能 } U &= \sum_i \epsilon_i n_i = \sum_{\epsilon} \epsilon e^{\alpha + \beta \epsilon} D(\epsilon) \\ &= \frac{4\sqrt{2} V \pi g m^{3/2}}{h^3} \int_0^\infty \frac{\epsilon^{3/2}}{e^{\alpha + \beta \epsilon} \pm 1} d\epsilon \end{aligned}$$

$$\frac{1}{e^{\alpha + \beta \epsilon} \pm 1} = e^{-\alpha - \beta \epsilon} \frac{1}{1 \pm e^{-\alpha - \beta \epsilon}} \approx e^{-\alpha - \beta \epsilon} (1 \mp e^{-\alpha - \beta \epsilon})$$

经典极限 (玻尔兹曼)
弱简并近似

$$N = \frac{4\sqrt{2} V \pi g m^{3/2}}{h^3} \int_0^\infty \frac{\epsilon^{1/2}}{e^{\alpha + \beta \epsilon} \pm 1} d\epsilon \approx \frac{4\sqrt{2} V \pi g m^{3/2}}{h^3} \int_0^\infty e^{-\alpha - \beta \epsilon} (1 \mp e^{-\alpha - \beta \epsilon}) d\epsilon$$

$$= \frac{4\sqrt{2} V \pi g m^{3/2}}{h^3} \left[\frac{1}{\beta} \int_0^\infty e^{-\beta \epsilon} \epsilon^{1/2} d\epsilon \mp \frac{1}{\beta} \int_0^\infty e^{-2\beta \epsilon} \epsilon^{1/2} d\epsilon \right]$$

$$= \frac{4\sqrt{2} V \pi g m^{3/2}}{h^3} \left[\frac{1}{\beta} \int_0^\infty e^{-\beta \epsilon} \epsilon^{1/2} d\epsilon \mp \frac{1}{2\beta} \int_0^\infty e^{-\beta \epsilon} \epsilon^{1/2} d\epsilon \right]$$

$$= \frac{2\sqrt{2} V \pi g m^{3/2}}{\beta h^3} \left(1 \mp \frac{1}{2} e^{-\alpha} \right)$$

$$U = \frac{4\sqrt{2} V \pi g m^{3/2}}{h^3} \int_0^\infty \frac{\epsilon^{3/2}}{e^{\alpha + \beta \epsilon} \pm 1} d\epsilon \approx \frac{4\sqrt{2} V \pi g m^{3/2}}{h^3} \int_0^\infty e^{-\alpha - \beta \epsilon} (1 \mp e^{-\alpha - \beta \epsilon}) \epsilon^{3/2} d\epsilon$$

$$= \frac{4\sqrt{2} V \pi g m^{3/2}}{h^3} \left[\frac{1}{\beta} \int_0^\infty e^{-\beta \epsilon} \epsilon^{3/2} d\epsilon \mp \frac{1}{2\beta} \int_0^\infty e^{-2\beta \epsilon} \epsilon^{3/2} d\epsilon \right]$$

$$= \frac{4\sqrt{2} V \pi g m^{3/2}}{h^3} \left[\frac{1}{\beta} \int_0^\infty e^{-\beta \epsilon} \epsilon^{3/2} d\epsilon \mp \frac{1}{4\beta} \int_0^\infty e^{-\beta \epsilon} \epsilon^{3/2} d\epsilon \right]$$

$$\frac{2\sqrt{2} V \pi g m^{3/2}}{\beta h^3} \left(1 \mp \frac{1}{4} e^{-\alpha} \right)$$

$$= e^{\alpha} \beta^{\frac{3}{2}} h^{\frac{3}{2}} \left(1 \mp \frac{1}{g} e^{-\alpha} \right)$$

$$\frac{h}{N} = \frac{3}{2} kT \left(1 \mp \frac{\sqrt{2}}{g} e^{-\alpha} \right) \approx \frac{3}{2} kT \left(1 \mp \frac{\sqrt{2}}{g} e^{-\alpha} \right) \left(1 \pm \frac{\sqrt{2}}{4} e^{-\alpha} \right)$$

$$\approx \frac{3}{2} kT \left(1 \pm \frac{\sqrt{2}}{g} e^{-\alpha} \right) \quad e^{\alpha} \approx \frac{V}{N} \left(\frac{2\pi m}{h^2} \right)^{3/2}$$

$$\Rightarrow u = \frac{3}{2} N kT \left[1 \pm \frac{\sqrt{2}}{g} \frac{V}{N} \left(\frac{h^2 N}{2\pi m} \right)^{3/2} \right] \quad \text{其中 } - \text{ 对应玻色气体, } + \text{ 对应费米气体.}$$

玻尔兹曼分布对应内能 量子统计关联 / 微观粒子全同性 引起附加内能。

§ 8.3 玻色-爱因斯坦凝聚

当 $e^{\beta\mu}$ 、 $n\lambda^3$ 超过一临界值 [2.612] 时, 粒子将在动量空间凝聚

化学势 μ 可由归一化条件求出:

$$N = \sum_i a_i = \sum_i \frac{1}{e^{\beta(\epsilon_i - \mu)} \pm 1} = \int_0^{\infty} \frac{1}{e^{\beta(\epsilon - \mu)} \pm 1} g(\epsilon) d\epsilon$$

$$= \frac{4\pi V}{h^3} m^{3/2} \int_0^{\infty} \frac{\epsilon^{1/2} d\epsilon}{e^{\beta(\epsilon - \mu)} \pm 1}$$

$$* a_i > 0 \rightarrow \epsilon > \mu$$

$$\hookrightarrow \mu \leq \epsilon_0 = 0 \text{ (最低能级)}$$

$$\Rightarrow \frac{4\pi V}{h^3} m^{3/2} \int_0^{\infty} \frac{\epsilon^{1/2} d\epsilon}{e^{\beta\epsilon} \pm 1} = N$$

对 $\mu = 0$ 的对应温度称为简并温度 T_c

$$\frac{4\pi V}{h^3} m^{3/2} \int_0^{\infty} \frac{\epsilon^{1/2} d\epsilon}{e^{\beta\epsilon} \pm 1} = N$$

$$\int_0^{\infty} \frac{x^{n-1} dx}{e^x \pm 1} \text{ (费米系统) 如何计算?}$$

$$A = \frac{2^{-1/2}}{\Gamma} \Gamma(3/2) \approx 2.612$$

$$= \begin{cases} \frac{4\pi V}{h^3} m^{3/2} (kT_c)^{3/2} \cdot \frac{\Gamma(3/2)}{2} \cdot 2.612 = 2.612 \left(\frac{2\pi m k T_c}{h^2} \right)^{3/2} & \text{玻色系统} \\ \frac{4\pi V}{h^3} m^{3/2} (kT_c)^{3/2} \cdot \frac{(2^{-1/2}) \Gamma(3/2)}{\Gamma} \cdot 2.612 = \frac{\Gamma(3/2)-1}{2} \cdot 2.612 \left(\frac{2\pi m k T_c}{h^2} \right)^{3/2} & \text{费米系统} \end{cases}$$

$$\Rightarrow T_c = \begin{cases} \left(\frac{N}{2.612} \right)^{2/3} \frac{h^2}{2\pi m k} & \text{玻色系统} \\ \left(\frac{N}{(2.612-1)} \right)^{2/3} \frac{h^2}{2\pi m k} & \text{费米系统} \end{cases}$$

$\mu \rightarrow 0$ 时, 归一化条件可写为

$$T^{3/2} \frac{4\pi V}{h^3} m^{3/2} k^{3/2} \int_0^{\infty} \frac{(\frac{\epsilon}{kT})^{1/2} d(\frac{\epsilon}{kT})}{e^{\frac{\epsilon}{kT}} \pm 1} = T^{3/2} \cdot A = N \quad (\text{常数})$$

$$\text{且 } T_c^{3/2} A = N$$

温度 T 与化学势 μ 为何? 比如理想气体 $\mu = -kT \ln e^{\beta}$
但对所有气体都适用吗?

$\Rightarrow T < T_c$ 时, $T^{3/2} A < N$, 需要修正!

修正: 在 $\epsilon=0$ 处, $\epsilon=0$ 的粒子被忽略

$$N = \sum_i a_i = \sum_i \frac{1}{e^{\beta(\epsilon_i - \mu)} \pm 1} = \int_0^{\infty} \frac{1}{e^{\beta(\epsilon - \mu)} \pm 1} g(\epsilon) d\epsilon$$

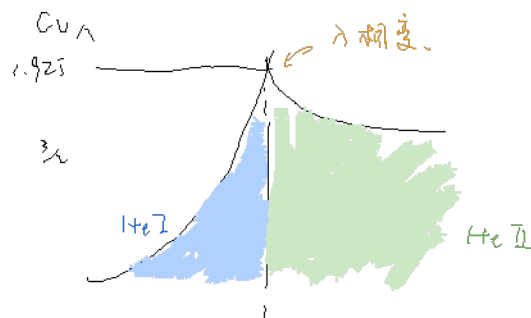
$$\begin{aligned}
 \text{其中 } n_{\varepsilon=0} &= \frac{4\sqrt{2} \lambda m^{3/2}}{h^3} \int_0^\infty \frac{\varepsilon^{1/2}}{e^{\frac{\varepsilon}{kT}} \pm 1} d\varepsilon = 0 \\
 &= \left(\frac{T}{T_c}\right)^{3/2} \frac{4\sqrt{2} \lambda m^{3/2}}{h^3} (kT_c)^{3/2} \int_0^\infty \frac{\left(\frac{\varepsilon}{kT_c}\right)^{1/2}}{e^{\frac{\varepsilon}{kT_c}} \pm 1} d\left(\frac{\varepsilon}{kT_c}\right) \\
 &= n \cdot \left(\frac{T}{T_c}\right)^{3/2}
 \end{aligned}$$

$$\Rightarrow n_0(T) = n \left[1 - \left(\frac{T}{T_c}\right)^{3/2}\right]$$

$T < T_c$ 时, 对于玻色气体, 大部份粒子将凝聚在 $\varepsilon=0$ 的基态

$$\begin{aligned}
 U &= \frac{4\sqrt{2} V \lambda m^{3/2}}{h^3} \int_0^\infty \frac{\varepsilon^{3/2}}{e^{\frac{\varepsilon}{kT}} - 1} d\varepsilon \\
 &= \frac{4\sqrt{2} V \lambda m^{3/2}}{h^3} (kT)^{3/2} \int_0^\infty \frac{\left(\frac{\varepsilon}{kT}\right)^{3/2}}{e^{\frac{\varepsilon}{kT}} - 1} d\left(\frac{\varepsilon}{kT}\right) \\
 &= \frac{4\sqrt{2} V \lambda m^{3/2}}{h^3} (kT)^{3/2} \cdot \frac{3\sqrt{\pi}}{4} \cdot 1.341 \\
 T_c &= \left(\frac{n}{2.612}\right)^{2/3} \frac{h^2}{2\pi m k} \\
 &= \frac{4\sqrt{2} V \lambda k T^{5/2}}{h^3} \cdot \frac{3\sqrt{\pi}}{4} \cdot 1.341 \cdot \left(\frac{n}{2.612}\right)^{2/3} \frac{h^2}{2\pi m k} \\
 &= \frac{4\sqrt{2} \cdot 3\sqrt{\pi} \cdot 1.341}{4 \cdot 2.612 \cdot (2\pi \lambda)^{3/2}} N k T \left(\frac{T}{T_c}\right)^{3/2} \\
 \approx 0.77 N k T \left(\frac{T}{T_c}\right)^{3/2}
 \end{aligned}$$

$$C_V = \left\langle \frac{\partial U}{\partial T} \right\rangle_V = 1.925 N k \left(\frac{T}{T_c}\right)^{3/2}$$



凝聚条件:

$$T < T_c = \left(\frac{n}{2.612}\right)^{2/3} \frac{h^2}{2\pi m k}$$

$$n \cdot \left(\frac{h^2}{2\pi m k T}\right)^{3/2} > 2.612$$

$$n \lambda^3 > 2.612$$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m\varepsilon}} \approx \frac{h}{\sqrt{2m\lambda k T}}$$

§ 8.4 光子气体

在平衡辐射问题中，光子数不守恒（壁在不断吸收/发射光子）

旧 - 化学势由 μ, V 守恒变为 $\mu = 0$ 守恒。

1. 粒子图像

$$\begin{cases} p = \hbar k \\ \varepsilon = \hbar \omega \end{cases} \xrightarrow{\omega = ck} \varepsilon = cp \quad \text{只有一个拉格朗日乘子 } \mu$$

对光子气体 $\mu = 0$?

$$a_\omega = \frac{1}{e^{\beta \varepsilon_\omega} - 1} \quad (\alpha = \frac{\mu}{kT} = 0 \rightarrow \mu = 0) \quad \text{光子是玻色子}$$

对光子 $g = 2$ (左右偏振)

$$D(p) dp = \frac{8\pi V}{h^3} p^2 dp \quad p = \frac{\varepsilon}{c} = \frac{\hbar \omega}{c}$$

$$D(\omega) d\omega = \frac{V}{\pi^2 c^3} \omega^2 d\omega = \omega^2 d\omega$$

$$a_\omega = \frac{V}{\pi^2 c^3} \frac{\omega^2 d\omega}{e^{\hbar \omega / kT} - 1}$$

$$\text{光子密度 } U(\omega) d\omega = a_\omega \hbar \omega = \frac{\hbar V}{\pi^2 c^3} \frac{\omega^3 d\omega}{e^{\hbar \omega / kT} - 1}$$

$$\approx \begin{cases} \frac{kT V}{\pi^2 c^3} \omega^2 d\omega \\ \frac{\hbar V}{\pi^2 c^3} \omega^3 e^{-\frac{\hbar \omega}{kT}} d\omega \end{cases}$$

低频 $\frac{\hbar \omega}{kT} \ll 1$

高频 $\frac{\hbar \omega}{kT} \gg 1$

2. 波动图像

辐射场 $\rightarrow$ 无穷多个单色平面波 (振幅由边界)

量子理论: 一个自由度的能量

$$\varepsilon_n = \hbar \omega (n + \frac{1}{2})$$

$$\text{光子数的平均值: } \bar{n} = \frac{1}{e^{\hbar \omega / kT} - 1} \quad (\text{粒子图像: 平均光子数})$$

3.

$$\begin{aligned} \text{内能: } U &= \int_0^\infty \frac{\hbar V}{\pi^2 c^3} \frac{\omega^3 d\omega}{e^{\hbar \omega / kT} - 1} = \frac{\hbar V}{\pi^2 c^3} \int_0^\infty \frac{\omega^3 d\omega}{e^{\hbar \omega / kT} - 1} \\ &= \frac{V k^4 T^4}{\pi^2 c^3 \hbar^3} \int_0^\infty \frac{x^3 dx}{e^x - 1} \quad x = \frac{\hbar \omega}{kT} \end{aligned}$$

$$= \frac{\pi^2 V k^4 T^4}{15 c^3 \hbar^3} \propto T^4$$

辐射场内能密度的极大值

$$\begin{aligned} \frac{d u(\omega)}{d \omega} &= \frac{\hbar V}{\pi^2 c^3} \frac{d}{d \omega} \left(\omega^3 e^{\frac{\hbar \omega}{kT} - 1} \right) \\ &= \frac{\hbar V}{\pi^2 c^3} \frac{1}{(e^{\frac{\hbar \omega}{kT} - 1})^2} \left[3 \omega^2 (e^{\frac{\hbar \omega}{kT} - 1}) - \omega^3 \frac{\hbar}{kT} e^{\frac{\hbar \omega}{kT}} \right] \\ &= \frac{\hbar V \omega^2}{\pi^2 c^3 (e^{\frac{\hbar \omega}{kT} - 1})^2} \left[3 - \frac{\hbar \omega}{kT} \right] e^{\frac{\hbar \omega}{kT} - 3} \end{aligned}$$

→ 维恩位移定律:

$$\omega_m = 2.822 \frac{kT}{\hbar}$$

4.

对于光子气体, 巨配分函数为

$$\begin{aligned} \ln \Xi &= \sum_i -\ln(1 - e^{-\alpha - \beta \epsilon_i}) \\ &= \int_0^\infty -\frac{V}{\pi^2 c^3} \omega^2 \ln(1 - e^{-\frac{\hbar \omega}{kT}}) d\omega \\ &= -\frac{V}{\pi^2 c^3} \left(\frac{kT}{\hbar} \right)^3 \int_0^\infty x^2 \ln(1 - e^{-x}) dx \\ &= -\frac{V}{\pi^2 c^3} \left(\frac{kT}{\hbar} \right)^3 \left[\frac{1}{3} x^3 \ln(1 - e^{-x}) \Big|_0^\infty - \frac{1}{3} \int_0^\infty \frac{x^3}{e^x - 1} dx \right] \\ &= \frac{\pi^2 V k^3 T^3}{45 c^3 \hbar^3} = \frac{\pi^2 V}{45 c^3 \hbar^3 \beta^3} \end{aligned}$$

$$\text{内能 } U = -\frac{\partial}{\partial \beta} \ln \Xi$$

$$= \frac{\pi^2 V k^4 T^4}{15 c^3 \hbar^3}$$

压强 (压强)

$$p = kT \frac{\partial}{\partial V} \ln \Xi = \frac{\pi^2 V k^4 T^4}{45 c^3 \hbar^3} = \frac{1}{3} \frac{U}{V}$$

$$\text{熵 } S = k \ln \Xi + \alpha \frac{\partial}{\partial \alpha} \ln \Xi + \beta \frac{\partial}{\partial \beta} \ln \Xi$$

$$= k (\ln \Xi - \beta \frac{\partial}{\partial \beta} \ln \Xi)$$

$$= k \left(\frac{\pi^2 V}{45 c^3 \hbar^3 \beta^3} - \beta \frac{\pi^2 V}{45 c^3 \hbar^3 \beta^3} \right)$$

$$= \frac{45 C^3 h^3 \beta^3}{45 C^3 h^3 \beta^3} = \frac{4\pi^2 V k^4 T^3}{45 C^3 h^3 \beta^3}$$

§ 8.5 金属中的自由电子

强简并条件 ($e^{\beta \epsilon} \ll 1$, $n \lambda^3 \gg 1$) 时费米子的状态.

1. 金属中电子气是否满足简并条件?

$$\text{电子气密度 } n = \frac{\rho}{m_0} = \frac{\rho N_A}{M_{\text{mol}}} \quad (\text{假设一个原子提供一个电子})$$

$$\lambda = \frac{h}{p} = \frac{h}{13m kT}$$

$$n \lambda^3 = \frac{\rho N_A h^3}{M_{\text{mol}}} \cdot \frac{1}{(13m kT)^3} \approx 6800 \quad \text{铜, } T=300\text{K}$$

2. 费米分布, 一个量子态上的平均量子数 ($\frac{a_i}{\omega_i}$)

$$f = \frac{1}{e^{\beta(\epsilon_i - \mu)} + 1}$$

电子自旋为 2, 态密度

$$D(\epsilon) d\epsilon = 2 \cdot \frac{2\pi V}{h^3} (2m)^{3/2} \epsilon^{1/2} d\epsilon = \omega_i$$

$$\begin{aligned} \Leftrightarrow \text{电子数 } a_i &= \frac{\omega_i}{e^{\beta(\epsilon_i - \mu)} + 1} \\ &= \frac{4\pi V (2m)^{3/2}}{h^3} \frac{\epsilon^{1/2} d\epsilon}{e^{\beta(\epsilon - \mu)} + 1} \end{aligned}$$

归一化条件:

$$\int_0^\infty \frac{4\pi V (2m)^{3/2}}{h^3} \frac{\epsilon^{1/2} d\epsilon}{e^{\beta(\epsilon - \mu)} + 1} = N \quad \rightarrow \text{可求 } \mu$$

3. $T=0\text{K}$ 时

$$f = \begin{cases} 1 & \epsilon < \mu(0) \\ 0 & \epsilon > \mu(0) \end{cases}$$

$$\begin{aligned} \Leftrightarrow N &= \int_0^{\mu(0)} \frac{4\pi V (2m)^{3/2}}{h^3} \epsilon^{1/2} d\epsilon \\ &= \frac{8\pi V (2m)^{3/2}}{3h^3} \mu^{3/2}(0) \end{aligned}$$

$$\rightarrow \text{费米能 } \epsilon_F = \mu(0) = \frac{h^2}{2m} \left(\frac{3N}{8\pi V} \right)^{2/3}$$

费米动量 $p_F = (2m\mu_0)^{1/2} = \hbar (8\lambda)^{3/2}$

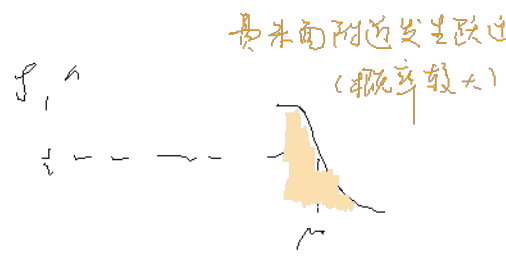
费米温度 $T_F = \mu_0 / k_B$ 铜 $8.1 \times 10^4 \text{ K}$

$$\begin{aligned} \text{内能 } U(\omega) &= \int_0^{\mu_0} \frac{4\pi V (2m)^{3/2}}{h^3} \xi^{3/2} d\xi \\ &= \frac{8\pi V (2m)^{3/2}}{5 h^3} \mu_0^{5/2} \\ &= \frac{3}{5} N \mu_0 \end{aligned}$$

压强 $p(\omega) = \frac{2}{3} \frac{U}{V} = \frac{2}{3} n \mu_0$ 铜 $3.8 \times 10^{10} \text{ Pa}$

4. $T > 0 \text{ K}$ 时

$$f \begin{cases} > \frac{1}{2} & \xi > \mu \\ = \frac{1}{2} & \xi = \mu \\ < \frac{1}{2} & \xi < \mu \end{cases}$$



只有能量在 $\mu \pm kT$ 附近电子才对热容有贡献

$N_{\text{eff}} = \frac{kT}{\mu} N$ — 个电子贡献 (取平均值)

$C_v = \frac{3}{2} k_B N_{\text{eff}} = \frac{3}{2} k_B N \frac{T}{T_F}$

总电子数 $N = \int_0^{\infty} \frac{4\pi V (2m)^{3/2}}{h^3} \xi^{3/2} \frac{1}{e^{\xi/\mu_0} + 1} d\xi$

内能 $U = \int_0^{\infty} \frac{4\pi V (2m)^{3/2}}{h^3} \xi^{5/2} \frac{1}{e^{\xi/\mu_0} + 1} d\xi$ $\xi = kT x + \mu_0$

$I = \int_0^{\infty} \frac{C e^d dx}{e^x + 1}$ ($C = \frac{4\pi V (2m)^{3/2}}{h^3}$)

$= \int_{-\mu_0/kT}^0 \frac{C (\mu_0 - kTx)^d}{e^x + 1} kT dx = \int_{-\mu_0/kT}^0 \frac{C (\mu_0 - kTx)^d}{e^x + 1} kT dx + \int_0^{\infty} \frac{C (\mu_0 + kTx)^d}{e^x + 1} kT dx$

$= \int_0^{\mu_0/kT} \frac{C (\mu_0 - kTx)^d}{e^{-x} + 1} kT dx + kT \int_0^{\infty} \frac{C (\mu_0 + kTx)^d}{e^x + 1} dx$ $\frac{1}{e^{-x} + 1} = \frac{e^x}{1 + e^x} = 1 - \frac{1}{e^x + 1}$

$= \int_0^{\mu_0/kT} C kT (\mu_0 - kTx)^d dx + kT \left[- \int_0^{\mu_0/kT} \frac{C (\mu_0 - kTx)^d}{e^x + 1} dx + \int_0^{\infty} \frac{C (\mu_0 + kTx)^d}{e^x + 1} dx \right]$
 因为 $\mu_0/kT \gg 1$ 且 x 较小的项贡献最大

$$= \int_0^{\mu} C (2\mu - \epsilon) d\epsilon + kT \int_0^{\infty} \frac{C}{e^{x+1}} \left[\underbrace{\mu + kTx}_{\text{在 } x=0 \text{ 展开}} - \underbrace{\mu - kTx}_{\text{在 } x=0 \text{ 展开}} \right] dx$$

$$= \int_0^{\mu} C \epsilon^2 d\epsilon + kT \int_0^{\infty} \frac{C}{e^{x+1}} 2\mu^{\alpha-1} kTx dx \quad \text{到 } -kT$$

$$= \int_0^{\mu} C \epsilon^2 d\epsilon + 2(kT)^2 C \mu^{\alpha-1} \int_0^{\infty} \frac{x dx}{e^{x+1}} = \frac{\lambda^2}{2}$$

$$= \int_0^{\mu} C \epsilon^2 d\epsilon + \frac{\lambda^2}{6} (kT)^2 C \mu^{\alpha-1}$$

$$I(\alpha = \frac{1}{2}) = \int_0^{\mu} C \epsilon^{\frac{1}{2}} d\epsilon + \frac{\lambda^2}{6} (kT)^2 C \frac{1}{2} \mu^{-\frac{1}{2}} = C \left[\frac{2}{3} \mu^{\frac{3}{2}} + \frac{\lambda^2}{12} (kT)^2 \frac{1}{\sqrt{\mu}} \right]$$

$$I(\alpha = \frac{3}{2}) = \int_0^{\mu} C \epsilon^{\frac{3}{2}} d\epsilon + \frac{\lambda^2}{6} (kT)^2 C \frac{3}{2} \mu^{\frac{1}{2}} = C \left[\frac{2}{5} \mu^{\frac{5}{2}} + \frac{\lambda^2}{4} (kT)^2 \sqrt{\mu} \right]$$

$$N = I(\alpha = \frac{1}{2}) = C \left[\frac{2}{3} \mu^{\frac{3}{2}} + \frac{\lambda^2}{12} (kT)^2 \frac{1}{\sqrt{\mu}} \right]$$

$$= \frac{2}{3} C \mu^{\frac{3}{2}} \left[1 + \underbrace{\frac{\lambda^2}{6} \left(\frac{kT}{\mu} \right)^2}_{T > 0 \text{ 修正项}} \right]$$

$$N = I(\alpha = \frac{3}{2}) = C \left[\frac{2}{5} \mu^{\frac{5}{2}} + \frac{\lambda^2}{4} (kT)^2 \sqrt{\mu} \right]$$

$$= \frac{2}{5} C \mu^{\frac{5}{2}} \left[1 + \underbrace{\frac{5}{2} \left(\frac{kT}{\mu} \right)^2}_{\text{修正项}} \right]$$